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# PAPERBOY

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*Samsung guided to 107.4 trillion won of quarterly operating profit — nine times last year's — and became the first Korean company to clear 100 trillion in a quarter.*

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**AI**

## OpenAI posts 722 machine-written mathematics papers, including a partial crack at Riemann

WHAT HAPPENED

OpenAI published [722 mathematics papers generated by an unreleased model](#), some posted to GitHub late Tuesday and the body of work surfacing Wednesday. They span roughly 20 subfields: more than 80 in theoretical computer science, three bounding how far matrix multiplication can be sped up, one introducing a new integer-multiplication algorithm, and more than a dozen on partial differential equations, among them a proof of a version of De Giorgi's conjecture and clarifications of several Navier-Stokes questions. On the Riemann hypothesis the model did not produce a complete proof but did prove the quasi-Riemann hypothesis. Many papers ship with Lean files, so a machine can check the proofs. The model is not named, no external mathematician is quoted, and no independent reviewers are identified — the 722 figure counts papers posted, not results verified.

WHAT IT MEANS

Lean files are the part that matters. A machine-checkable proof does not need a referee to believe it, which is the first time an AI claim about mathematics has come with its own audit trail. The framing is still OpenAI's: "discoveries" is doing work that peer review has not done. Treat it as 722 submissions.

WATCH NEXT

Whether named mathematicians verify any of the PDE proofs, and whether OpenAI ships Lean files for the rest.

# SpaceX reported to be raising \$40bn in debt to buy Nvidia chips, with Apollo leading

## WHAT HAPPENED

SpaceX is [reported to be seeking about \\$40bn](#) to fund an Nvidia chip order, split roughly \$10bn in bank loans and \$30bn in investment-grade debt, with Apollo Global Management expected to lead and place the paper, and Pimco among a small group of lenders in talks. The deal is expected to close in 2027. The report originates with the Financial Times; every other outlet carrying it traces back to that one story, and no named party — SpaceX, Apollo, Pimco or Nvidia — has said anything on the record. Musk has said SpaceX will use Nvidia hardware exclusively in its data centres and said last month that the Colossus 2 facility could more than double its Nvidia chip count by December. SpaceX shares fell about 2% in early trade Wednesday on the report.

## WHAT IT MEANS

The novelty is the structure, not the size. Thirty billion of investment-grade paper means a rating agency has to put a letter on a chip purchase by a company whose main business is rockets. That is a bond market being asked to underwrite depreciation risk on GPUs — the thing nobody in this build-out has yet priced.

## WATCH NEXT

A rating, a prospectus, or an on-the-record confirmation from Apollo. Until one appears this is one newspaper's reporting.

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# Samsung guides to 107.4 trillion won, nine times last year, and a 55% operating margin

## WHAT HAPPENED

Samsung Electronics filed [Q3 2026 earnings guidance](#) on Thursday: consolidated operating profit of about 107.40tn won and sales of about 195tn won, K-IFRS. Both headline numbers are medians — Korean disclosure rules bar ranged estimates, so the release gives sales of 194-196tn and operating profit of 107.3-107.5tn and says the stated figures are the midpoints. Against the comparison periods in the same release — Q2 2026 at 171.5tn sales and 89.49tn operating profit, Q3 2025 at 86.06tn and 12.17tn — that is operating profit up about 783% year on year and 20% on the quarter, on sales up 127%. Seoul Economic Daily puts the FnGuide three-month consensus at 200.28tn revenue and 106tn operating profit, so profit beat and revenue missed. The implied operating margin, 55.1% against 52.2% in Q2, is arithmetic off the two disclosed lines, not a company figure. The release carries no segment breakdown and no net income, so nothing circulating today about the memory division is sourced to Samsung.

## WHAT IT MEANS

This is the memory supercycle arriving in a single number. A 55% operating margin is not a semiconductor margin; it is a toll-booth margin, and it says Samsung is pricing DRAM and HBM into a market with no alternative supplier. The revenue miss against a profit beat is the tell — volumes are not what did this, price is.

## WATCH NEXT

The full results late in October for the semiconductor segment split, and whether any phone or laptop maker finally puts a memory surcharge in writing.

# TSMC files September revenue early: NT\$511.9bn, up 54.6%, and flat on the month

## WHAT HAPPENED

TSMC's investor site now shows [September 2026 consolidated net revenue of NT\\$511,857 million, up 54.6% year on year](#), filed ahead of its usual mid-month cadence. That is marginally below August's NT\$514,806 million, which was up 53.3% — a sequential decline of about 0.6%, a figure derived from the two filed months rather than reported, since the page publishes no month-on-month line. The 2026 figures are flagged unaudited and October through December are blank. Two consecutive months above NT\$510bn with growth still above 50% is the cleanest public read on AI-accelerator foundry demand going into the fourth quarter.

## WHAT IT MEANS

Flat beats falling, and at this scale flat is the point: TSMC is running against a capacity ceiling, not a demand one. Growth above 50% a year into the build-out means the comparison base has stopped flattering the number. Watch for the month the sequential line turns genuinely negative, because that will be the first honest signal of a top.

## WATCH NEXT

Q3 results on 15 October, where capex guidance and the advanced-node mix matter more than the monthly line.

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## September minutes: a unanimous hike, a divided rationale, most expecting another

### WHAT HAPPENED

The [minutes of the 15-16 September meeting](#) show the Committee raised the target range 25 basis points (a quarter of a percentage point) to 3.75-4.00% on a 12-0 vote. The split was in reasoning. "Most participants assessed that another increase in the target range for the federal funds rate would likely be appropriate by year end." Several "viewed the current policy rate as not restrictive or only mildly restrictive"; many warned that "the longer energy prices remained elevated, the greater the risk that cost increases in certain sectors could lead to broader price pressures". Staff had August PCE at 3.8% and unemployment at 4.1%. The [dot plot](#) puts 16 of 18 participants above the current range.

### WHAT IT MEANS

A unanimous vote with that much disagreement under it is a committee holding together by moving. The energy passage is the live wire: Kevin Warsh's Fed is treating an oil shock as a reason to tighten rather than look through, the opposite of the 2022 playbook.

### WATCH NEXT

September CPI on 14 October and the 27-28 October meeting.

## Records give way as the long end backs up and dealers take 2.5% of a \$39bn 10-year

### WHAT HAPPENED

No index set a record. The [S and P 500 closed at 7,801.77, down 0.22%](#), the Dow at 51,179.87 (-0.66%), the Nasdaq Composite at 27,538.69 (-0.22%) and the Russell 2000 at 2,793.20 (-1.31%). [Treasury's par yields](#) went 2-year 4.77%, 10-year 5.28%, 30-year 5.67%, against 4.79, 5.27 and 5.64 on Tuesday, with the 10-year above 5.35% intraday, a 24-year high. The [\\$39bn 10-year reopening](#) stopped at 5.300%, bid-to-cover 2.77; of \$38.67bn competitive accepted, primary dealers took \$984m, or 2.5%.

### WHAT IT MEANS

A 2.5% dealer take-down is near a record low and the most bullish thing in the release — dealers absorb only what nobody else wants, and at 5.30% buyers took all of it. But real money is buying the belly while the 30-year backs up, steepening the curve into an unfinished auction calendar.

### WATCH NEXT

Thursday's \$22bn 30-year reopening, and whether the 10-year holds 5.25% into CPI.

# Oil slipped despite a 3.2 million barrel crude draw; Cenovus buys Athabasca for \$4bn

## WHAT HAPPENED

Crude fell Wednesday: November WTI settled at \$88.28, down 1.30%, and December Brent at \$100.20, down 0.38%, from Tuesday's \$89.44 and \$100.58. It fell against a bullish print — the EIA's report for the week to 2 October had commercial crude down 3,186,000 barrels where the market looked for a build of about 1,722,000 — one outlet's capture of the release. The reserve stood at 283.8 million barrels on 25 September, the most recent level available. Separately, [Cenovus agreed to buy Athabasca for \\$4bn](#).

## WHAT IT MEANS

A three-million-barrel draw that cannot lift the price tells you the bid is geopolitical, not physical. Brent at \$100 is holding on Hormuz risk, and risk premia deflate on no news at all — while the reserve is spent capping a price inventories say is already supported.

## WATCH NEXT

Next Wednesday's EIA print and whether Brent holds \$100 without a headline.

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## STARTUPS

# Parallel Systems raises more than \$100m for autonomous electric freight rail

## WHAT HAPPENED

Parallel Systems, founded by SpaceX alumni, [closed a Series C of more than \\$100m led by AVP](#), with Hillspire, Agility Global and Cobalt Capital new and four existing backers following on. It builds self-propelled battery-electric autonomous rail vehicles — the third generation is called Panther — and runs an FRA-supervised pilot in Georgia with short-line operator Genesee and Wyoming. Founder and chief executive Matt Soule is putting the money into commercialisation and manufacturing. Lifetime funding is now above \$200m, cumulative rather than this round, and no valuation was disclosed.

## WHAT IT MEANS

Rail is the one freight mode where autonomy needs no new regulatory category: the track is private and the FRA already supervises unmanned movements. The hard part is persuading Class I railroads to let someone else's hardware onto their network. Starting with a short line is the right wedge and the slow one.

## WATCH NEXT

Whether a Class I railroad signs, and whether the pilot converts to revenue service.

# General Medicine takes \$120m from a16z and Eli Lilly for a price-visible health catalogue

## WHAT HAPPENED

General Medicine [raised a \\$120m Series B led by Andreessen Horowitz](#), announced Tuesday, with Matrix, VXI Capital, Eli Lilly and Company, Mercy Health and BoxGroup. The San Francisco company runs what it calls a healthcare store: one catalogue of medications, lab tests, telehealth and specialist e-consults, with cash and insurance prices side by side. Chief executive TJ Parker and president Elliot Cohen founded PillPack. Total funding is \$152m, cumulative rather than a round size; no valuation disclosed.

## WHAT IT MEANS

Putting a cash price next to an insurance price is the whole product, and a bet that patients will shop once they can see. Lilly on the cap table is the odd signature — a manufacturer investing in the shelf it sells from. The risk is the one that has killed every price-transparency startup: the people who set prices benefit from their invisibility.

## WATCH NEXT

Whether a large insurer or pharmacy benefit manager agrees to list.

# Sriram Krishnan reported to be raising \$500m for a national-security AI fund

## WHAT HAPPENED

Sriram Krishnan, who stepped down in June as the White House's senior policy adviser on artificial intelligence, is [seeking around \\$500m for a new venture fund](#), Axios reported Wednesday, targeting growth-stage AI companies relevant to US or allied national security. The vehicle is early: no name, no announced limited partners, no team. Krishnan was an Andreessen Horowitz partner before the administration, and an AI institute he floated after leaving may fold into the fund. Axios says only that it "has learned" of the raise, with no named sources and no comment from Krishnan — treat the number as unconfirmed.

## WHAT IT MEANS

The corridor between AI policy and AI capital now has a fund at the end of it. Half a billion is small against the compute rounds and large against defence-adjacent software, and the pitch is access: he helped write the rules his portfolio will clear.

## WATCH NEXT

A filed Form D, a named anchor investor, or a first deal.

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## Axios reports the Pentagon told CENTCOM to be ready to resume major operations against Iran

### WHAT HAPPENED

The Pentagon told US Central Command "several days ago" to complete preparations to resume major combat operations against Iran — bombing of energy, infrastructure and nuclear targets, likely alongside Israel — [Axios reported Wednesday](#). No date is set and no decision has been made. Sources said strikes could come before the midterms, and that a resumption in three to four weeks risks electoral damage via oil prices. The national security team met at Camp David on Friday 2 October. Rubio, the only named source, said the team "constantly are meeting on long-term strategic planning" but added "I wouldn't call it routine".

### WHAT IT MEANS

One reporter's scoop, carried in two outlets under the same byline — reporting, not fact. It is worth printing because the market is already trading it: Brent above \$100 on building inventories is a war premium.

### WATCH NEXT

A congressional notification, a carrier movement, or an on-the-record statement.

## Judge presses the FCC for two hours over its early review of ABC's licences, and rules on nothing

### WHAT HAPPENED

Judge Loren L. AliKhan heard two hours of argument Tuesday in [American Broadcasting Companies, Inc. v. FCC](#), No. 1:26-cv-02902 (D.D.C.). ABC, Disney and eight ABC stations seek a restraining order and preliminary injunction — not vacatur — against the early call-in of those licences, ordered by Media Bureau directive [DA 26-416](#) of 28 April as "essential to the proper conduct of an investigation". ABC says the threat is already shaping its editorial choices; the government called it a routine regulatory inquiry. The FCC has moved separately to dismiss, arguing any harm to ABC's speech is "self-inflicted". AliKhan ruled on nothing.

### WHAT IT MEANS

Licence renewal is the only real leverage a regulator has over a network, and pulling it forward by years is the point: the process is the penalty. The self-inflicted argument is the striking part — if the investigation changes what ABC broadcasts, the FCC says that is ABC's choice.

### WATCH NEXT

A ruling on the restraining order, due any day.



# The diesel order has still produced nothing, and the AI czar still has no charter

## WHAT HAPPENED

The [presidential actions page](#) carries two proclamations dated 7 October and nothing else. The 5 October executive order on emergency diesel tax relief remains the latest order, and three days on there is no Treasury determination, no IRS guidance and no Federal Register citation; the five-day clock runs out around Saturday. Its only effect remains red-dyed diesel on highways to 31 December. The 29 September super-intelligence order is still the only AI instrument: nothing establishes a Super Intelligence Force or names Jay Clayton to run it — a fifth day with a czar and no legal basis for him.

## WHAT IT MEANS

Two vacuums. The diesel relief does not exist until Treasury says it is lawful, and a reporting clock cannot start without an instrument. Announcement as policy: the press treats the statement as the decision while the agencies hold nothing.

## WATCH NEXT

A Treasury determination before the clock lapses, and any AI charter.

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## WORLD POLITICS

# Britain renames its East Jerusalem consulate rather than close it

## WHAT HAPPENED

Israel's 30-day deadline for Britain to shut its East Jerusalem consulate general, retaliation for UK sanctions on West Bank settlements, expired Thursday. Britain renamed the post. Foreign Secretary Ed Miliband [said](#) "our Consulate General building will not be closing, and will be known as the UK Mission, Jerusalem from today", and that Britain "will not take reciprocal action while such discussions are underway and credible". It rejects Israel's legal right to close the post. Gideon Saar, Israel's foreign minister, said seven diplomats may remain and 20, the consul general among them, are going — his figures alone, confirmed by nobody in London. Britain expelled nobody.

## WHAT IT MEANS

Both governments wrote themselves an exit: Britain keeps the legal position while conceding the function, Israel keeps the win while leaving seven people in place. The Sheikh Jarrah mission is how Britain talks to Palestinians, so the nameplate matters less than the channel does.

## WATCH NEXT

Whether the seven are accredited, and whether talks resume.

# Eritrean troops reported inside Tigray, and Asmara does not deny it

## WHAT HAPPENED

At least two witnesses told Reuters they saw dozens of Eritrean soldiers drive through Adigrat, 25km south of the border, in two trucks on Wednesday, [in off-white camouflage](#). Gebreigziabher Beyene, who commands the pro-government Tigray Peace Force, said "the Eritrean army has entered Tigray in uniform, equipped with heavy weaponry" — a belligerent describing his own fight, not proof. Eritrean Information Minister Yemane Meskel did not deny a crossing, writing on X, as quoted by Al Jazeera, that Ethiopia was constructing "a fallacious narrative of 'self-defense'". Reuters could not verify the sightings. It would be the first since fighting resumed on 23 September.

## WHAT IT MEANS

A non-denial from Asmara is the loudest thing here. Eritrean regulars inside Tigray turn a civil conflict into an interstate one, and it is the move that in 2020 produced the war's worst atrocities. Two weeks after Mekelle changed hands, nobody governs it on the record.

## WATCH NEXT

Geolocated footage of the convoy, and a named Tigrayan interim authority.

# Spain lifts the warrants against Puigdemont without ruling on the amnesty

## WHAT HAPPENED

Supreme Court magistrate Pablo Llarena on Tuesday set aside the arrest warrants against Carles Puigdemont, Antoni Comin and Lluís Puig, effective immediately and until a new express ruling. He was explicit that it does not imply applying the Catalan amnesty: he had not received the court's final text and judged lifting them proportionate, citing the right to liberty against an unjustified detention. The trigger was a [Constitutional Court ruling the same day](#), carried 7-5, upholding an appeal by former minister Dolors Bassa and ordering the criminal chamber to annul the orders refusing her the amnesty. The five dissenters accused the majority of supplanting the Supreme Court.

## WHAT IT MEANS

This clears the obstacle to Puigdemont returning after nearly nine years abroad while settling none of the law. Llarena removed the warrants and kept the prosecution, handing him a decision rather than a guarantee. A 7-5 court is no stable foundation for it.

## WATCH NEXT

Whether Puigdemont names a return date.

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## SOURCES

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Federal Reserve — FOMC minutes, 15-16 September 2026

Federal Reserve — September 2026 projections table

Treasury — daily yield curve, October 2026

Treasury FiscalData — auction results, 7 October 2026

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Investing.com — WTI crude futures, historical settles

Investing.com — Brent crude futures, historical settles

Rigzone — Cenovus to buy Athabasca

SiliconANGLE — OpenAI publishes 722 AI-generated maths papers

Benzinga — SpaceX, Apollo and a \$40bn Nvidia chip financing

Parallel Systems — \$100m Series C

General Medicine — \$120m Series B

Axios — Sriram Krishnan's national-security AI fund

Axios — Pentagon tells CENTCOM to prepare for Iran operations

Clearinghouse — ABC v. FCC, No. 1:26-cv-02902 (D.D.C.)

FCC — Media Bureau directive DA 26-416

JURIST — judge hears ABC's challenge to the FCC licence review

The White House — presidential actions

FCDO — Foreign Secretary statement on the UK's presence in Jerusalem

Al Jazeera — UK consulate signage removed as the Israeli deadline passes

Al Jazeera — witnesses spot Eritrean troops inside northern Ethiopia

Catalan News — Spain lifts the arrest warrants against Puigdemont